In [1]:
```
#import necessary packages
import pandas as pd
import numpy as np
import openpyxl
```

In [2]:
```
#read in data file
data = pd.read_csv('initial_variables.csv')
```

In [3]:
```
#get a quick view of the data
data.head()
```

Out[3]:

	_AGEG5YR	ALCDAY4	AVEDRNK4	_BMI5	_BMI5CAT	CHECKUP1	CHILDREN	_CURECI3
0	12	888.0	NaN	2249.0	2.0	1.0	88.0	1
1	13	888.0	NaN	2583.0	3.0	1.0	88.0	1
2	8	230.0	2.0	2253.0	2.0	4.0	88.0	1
3	13	888.0	NaN	2509.0	3.0	1.0	88.0	1
4	6	888.0	NaN	1977.0	2.0	1.0	88.0	1

5 rows × 54 columns

In [4]:
```
#remove columns deemed unnecessary
data=data.drop(columns=["ALCDAY4","_BMI5CAT","CVDCRHD4","ECIGNOW3","_LCSAGE","_LCSC
                        "_LCSSMKG","_LCSYQTS","MENTCIGS","MENTECIG","_MENT14D","_PAC
                        "_RFSMOK3","SMOKE100","SMOKDAY2","_SMOKER3","STOPSMK2","_TOT
```

In [5]:
```
#Recategorize into 3 groups + unknown
conditions = [
    (data['_AGEG5YR'] < 5),
    (data['_AGEG5YR'] >= 5) & (data['_AGEG5YR'] < 9),
    (data['_AGEG5YR'] >= 9) & (data['_AGEG5YR'] < 14),
    (data['_AGEG5YR'] == 14)
]
choices = ['Young_Adult', 'Middle_Aged', 'Elderly', "Unknown"]

data['Age_Category'] = np.select(conditions, choices, default='Unknown')
data['Age_Category']=pd.Categorical(data.Age_Category)
```

In [6]:
```
#Redefine target variable as binary
data['_CURECI3']=pd.to_numeric(data._CURECI3)
conditions = [
    (data["_CURECI3"] == 1),
    (data["_CURECI3"] == 2),
    (data["_CURECI3"] == 9)
]
choices = [0, 1, np.nan]
```

```
data["E_Cig_User"] = np.select(conditions, choices, default=np.nan)
data['E_Cig_User']=pd.Categorical(data.E_Cig_User)
```

In [7]:
```
#Rename column to better name
data=data.rename(columns={"_BMI5":"BMI"})
```

In [8]:
```
#Recategorize into better groups
conditions = [
    (data["USENOW3"] == 1),
    (data["USENOW3"] == 2),
    (data["USENOW3"] == 3),
    (data["USENOW3"] == 7) | (data["USENOW3"] == 9)
]
choices = ["Every Day", "Some days", "Not at all", "Unknown"]

data["Other_Tobacco_Usage"] = np.select(conditions, choices, default="Unknown")
data['Other_Tobacco_Usage']=pd.Categorical(data.Other_Tobacco_Usage)
```

In [9]:
```
#Recategorize into better groups
conditions = [
    (data["SEXVAR"] == 1),
    (data["SEXVAR"] == 2)
]
choices = ["Male", "Female"]

data["Sex"] = np.select(conditions, choices, default="Unknown")
data['Sex']=pd.Categorical(data.Sex)
```

In [10]:
```
#Recategorize into better groups
conditions = [
    (data["_RACE"] == 1),
    (data["_RACE"] == 2),
    (data["_RACE"] == 3),
    (data["_RACE"] == 4),
    (data["_RACE"] == 5),
    (data["_RACE"] == 6),
    (data["_RACE"] == 7),
    (data["_RACE"] == 8),
    (data["_RACE"] == 9),
]
choices = ["White", "Black", "American Indian or Alaskan Native", "Asian",
           "Native American or other Pacific Islander", "Other Race", "Multiracial"
           "Hispanic", "Unknown"]

data["Race"] = np.select(conditions, choices, default="Unknown")
data['Race']=pd.Categorical(data.Race)
```

In [11]:
```
#Make these variable observations unknown since this is a continuous variable. Rena
data["AVEDRNK4"]=data["AVEDRNK4"].replace(88,0)
data["AVEDRNK4"]=data["AVEDRNK4"].replace(77,np.nan)
data["AVEDRNK4"]=data["AVEDRNK4"].replace(99,np.nan)
data=data.rename(columns={"AVEDRNK4":"Average_Number_of_Drinks_30_Days"})
```

In [12]:
```
#Recategorize into better groups
conditions = [
    (data["CHECKUP1"] == 1),
    (data["CHECKUP1"] == 2),
    (data["CHECKUP1"] == 3),
    (data["CHECKUP1"] == 4),
    (data["CHECKUP1"] == 7),
    (data["CHECKUP1"] == 8),
    (data["CHECKUP1"] == 29)
]
choices = ["Within_the_last_year", "Within_the_past_two_years", "Within_the_past_fi
          "Unknown", "Never", "Unknown"]

data["Time_Since_Last_Check_Up"] = np.select(conditions, choices, default="Unknown"
data['Time_Since_Last_Check_Up']=pd.Categorical(data.Time_Since_Last_Check_Up)
```

In [13]:
```
#Make these variable observations unknown since this is a continuous variable. Rena
data["CHILDREN"]=data["CHILDREN"].replace(88,0)
data["CHILDREN"]=data["CHILDREN"].replace(99,np.nan)
data=data.rename(columns={"CHILDREN":"Number_of_Children"})
```

In [14]:
```
#Redefine variable as binary
data["_DENVST3"]=data["_DENVST3"].replace(2,0)
data["_DENVST3"]=data["_DENVST3"].replace(9,np.nan)
data=data.rename(columns={"_DENVST3":"Dentist_Visit_Last_Year"})
```

In [15]:
```
#Recategorize into better groups
conditions = [
    (data["EXERANY2"] == 1),
    (data["EXERANY2"] == 2),
    (data["EXERANY2"] == 7),
    (data["EXERANY2"] == 9)
]
choices = [1, 0, np.nan, np.nan]

data["Exercise_Outside_Work_Past_Month"] = np.select(conditions, choices, default=n
```

In [16]:
```
#Redefine variable as binary
data["_HLTHPL2"]=data["_HLTHPL2"].replace(2,0)
data["_HLTHPL2"]=data["_HLTHPL2"].replace(9,np.nan)
data=data.rename(columns={"_HLTHPL2":"Health_Insurance"})
```

In [17]:
```
#Make these variables unknown since this is a continuous variable. Rename to a bett
data["MARIJAN1"]=data["MARIJAN1"].replace(88,0)
data["MARIJAN1"]=data["MARIJAN1"].replace(77,np.nan)
data["MARIJAN1"]=data["MARIJAN1"].replace(99,np.nan)
data=data.rename(columns={"MARIJAN1":"Marijuana_Use_30_Days"})
```

In [18]:
```
#Make these variable observations unknown since this is a continuous variable. Rena
data["MEDCOST1"]=data["MEDCOST1"].replace(2,0)
data["MEDCOST1"]=data["MEDCOST1"].replace(7,np.nan)
data["MEDCOST1"]=data["MEDCOST1"].replace(9,np.nan)
data=data.rename(columns={"MEDCOST1":"Medical_Visit_Missed_Due_to_Cost"})
```

In [19]:
```
#Recategorize into better groups
conditions = [
    (data["SDLONELY"] == 1),
    (data["SDLONELY"] == 2),
    (data["SDLONELY"] == 3),
    (data["SDLONELY"] == 4),
    (data["SDLONELY"] == 5),
    (data["SDLONELY"] == 7),
    (data["SDLONELY"] == 9)
]
choices = ["Always", "Usually", "Sometimes", "Rarely", "Never", "Unknown", "Unknown

data["Loneliness"] = np.select(conditions, choices, default= "Unknown")
```

In [20]:
```
#Recategorize into better groups
conditions = [
    (data["_RFHLTH"] == 1),
    (data["_RFHLTH"] == 2),
    (data["_RFHLTH"] == 9),
]
choices = ["Good_or_Better_Health", "Fair_or_Poor_Health", "Unknown"]

data["Health_Rating"] = np.select(conditions, choices, default= "Unknown")
```

In [21]:
```
#Recategorize into better groups
conditions = [
    (data["PREDIAB2"] == 1),
    (data["PREDIAB2"] == 2),
    (data["PREDIAB2"] == 3),
    (data["PREDIAB2"] == 7),
    (data["PREDIAB2"] == 9)
]
choices = ["Yes", "Yes_during_Pregnancy", "No", "Unknown", "Unknown"]

data["Prediabetic_Diagnosis"] = np.select(conditions, choices, default= "Unknown")
```

In [22]:
```
#Make these variable observations unknown since this is a continuous variable. Rena
data["MENTHLTH"]=data["MENTHLTH"].replace(88,0)
data["MENTHLTH"]=data["MENTHLTH"].replace(77,np.nan)
data["MENTHLTH"]=data["MENTHLTH"].replace(99,np.nan)
data=data.rename(columns={"MENTHLTH":"Poor_Mental_Health_Days_Last_Month"})
```

In [23]:
```
#Recategorize into better groups
conditions = [
    (data["INCOME3"] == 1),
    (data["INCOME3"] == 2),
    (data["INCOME3"] == 3),
    (data["INCOME3"] == 4),
    (data["INCOME3"] == 5),
    (data["INCOME3"] == 6),
    (data["INCOME3"] == 7),
    (data["INCOME3"] == 8),
    (data["INCOME3"] == 9),
    (data["INCOME3"] == 10),
```

```
    (data["INCOME3"] == 11),
    (data["INCOME3"] == 77),
    (data["INCOME3"] == 99)
]
choices = ["Less_than_$10,000", "Less_than_$15,000", "Less_than_$20,000",
           "Less_than_$25,000", "Less_than_$35,000", "Less_than_$50,000",
           "Less_than_$75,000", "Less_than_$100,000", "Less_than_$150,000",
           "Less_than_$200,000", "$200,000+", "Unknown", "Unknown"]

data["Annual_Income"] = np.select(conditions, choices, default= "Unknown")
```

In [24]:
```
#Recategorize into better groups
conditions = [
    (data["_PHYS14D"] == 1),
    (data["_PHYS14D"] == 2),
    (data["_PHYS14D"] == 3),
    (data["_PHYS14D"] == 9)
]
choices = ["0_Days", "1-13_Days", "14+_Days", "Unknown"]

data["Days_Poor_Physical_Health"] = np.select(conditions, choices, default= "Unknow
```

In [25]:
```
#Make these variable observations unknown since this is a continuous variable. Rena
data["PHYSHLTH"]=data["PHYSHLTH"].replace(88,0)
data["PHYSHLTH"]=data["PHYSHLTH"].replace(77,np.nan)
data["PHYSHLTH"]=data["PHYSHLTH"].replace(99,np.nan)
data=data.rename(columns={"PHYSHLTH":"Poor_Physical_Health_Days_Last_Month"})
```

In [26]:
```
#Recategorize into better groups
conditions = [
    (data["EDUCA"] == 1),
    (data["EDUCA"] == 2),
    (data["EDUCA"] == 3),
    (data["EDUCA"] == 4),
    (data["EDUCA"] == 5),
    (data["EDUCA"] == 6),
    (data["EDUCA"] == 9)
]
choices = ["Never_attended_school_or_only_Kindergarten", "Grades_1-8_(elementary)",
           "Grades_9-11_(some_high_school)", "Grade_12_or_GED_(high_school_graduate"
           "College_1-3_years_(some_college_or_technical_school", "College_4_years_o
           "Unknown"]

data["Education"] = np.select(conditions, choices, default= "Unknown")
```

In [27]:
```
#Recategorize into better groups
conditions = [
    (data["EMTSUPRT"] == 1),
    (data["EMTSUPRT"] == 2),
    (data["EMTSUPRT"] == 3),
    (data["EMTSUPRT"] == 4),
    (data["EMTSUPRT"] == 5),
    (data["EMTSUPRT"] == 7),
    (data["EMTSUPRT"] == 9)
]
```

```
choices = ["Always","Usually","Sometimes","Rarely","Never", "Unknown","Unknown"]

data["Social_Emotional_Support"] = np.select(conditions, choices, default= "Unknown
```

In [28]:
```
#Recategorize into better groups
conditions = [
    (data["GENHLTH"] == 1),
    (data["GENHLTH"] == 2),
    (data["GENHLTH"] == 3),
    (data["GENHLTH"] == 4),
    (data["GENHLTH"] == 5),
    (data["GENHLTH"] == 7),
    (data["GENHLTH"] == 9)
]
choices = ["Excellent", "Very_good", "Good", "Fair", "Poor", "Unknown","Unknown"]

data["General_Health_Personal_Rating"] = np.select(conditions, choices, default= "U
```

In [29]:
```
#Recategorize into better groups
conditions = [
    (data["LSATISFY"] == 1),
    (data["LSATISFY"] == 2),
    (data["LSATISFY"] == 3),
    (data["LSATISFY"] == 4),
    (data["LSATISFY"] == 7),
    (data["LSATISFY"] == 9)
]
choices = ["Very_satisfied", "Satisfied", "Dissatisfied", "Very_dissatified", "Unkn

data["Life_Satisfaction"] = np.select(conditions, choices, default= "Unknown")
```

In [30]:
```
#Recategorize into better groups
conditions = [
    (data["MARITAL"] == 1),
    (data["MARITAL"] == 2),
    (data["MARITAL"] == 3),
    (data["MARITAL"] == 4),
    (data["MARITAL"] == 5),
    (data["MARITAL"] == 6),
    (data["MARITAL"] == 9)
]
choices = ["Married", "Divorced", "Widowed", "Separated", "Never_Married", "Member_
          "Unknown"]

data["Marital_Status"] = np.select(conditions, choices, default= "Unknown")
```

In [31]:
```
#Split initial variable into 3 variables: day, week, and month
conditions = [
    (data["SSBSUGR2"] < 200)
]
choices = [data["SSBSUGR2"]-100]

data["Sugar_Soda_Drinks_Per_Day"] = np.select(conditions, choices, default= np.nan)
```

In [32]:
```
#Split initial variable into 3 variables: day, week, and month
conditions = [
    (data["SSBSUGR2"] < 300)
]
choices = [data["SSBSUGR2"]-200]

data["Sugar_Soda_Drinks_Per_Week"] = np.select(conditions, choices, default= np.nan
```

In [33]:
```
#Split initial variable into 3 variables: day, week, and month
conditions = [
    (data["SSBSUGR2"] < 400)
]
choices = [data["SSBSUGR2"]-300]

data["Sugar_Soda_Drinks_Per_Month"] = np.select(conditions, choices, default= np.na
```

In [34]:
```
#Split initial variable into 3 variables: day, week, and month
conditions = [
    (data["SSBFRUT3"] < 200)
]
choices = [data["SSBFRUT3"]-100]

data["Non_Soda_Sugar_Drinks_Per_Day"] = np.select(conditions, choices, default= np.
```

In [35]:
```
#Split initial variable into 3 variables: day, week, and month
conditions = [
    (data["SSBFRUT3"] < 300)
]
choices = [data["SSBFRUT3"]-200]

data["Non_Soda_Sugar_Drinks_Per_Week"] = np.select(conditions, choices, default= np
```

In [36]:
```
#Split initial variable into 3 variables: day, week, and month
conditions = [
    (data["SSBFRUT3"] < 400)
]
choices = [data["SSBFRUT3"]-300]

data["Non_Soda_Sugar_Drinks_Per_Month"] = np.select(conditions, choices, default= n
```

In [37]:
```
#Dropping columns that are unnecessary since new variables have been created
data=data.drop(columns=['_AGEG5YR','_CURECI3','USENOW3','SEXVAR',"_RACE","CHECKUP1"
                        "SDLONELY","_RFHLTH","PREDIAB2","INCOME3","_PHYS14D","EDUCA"
                        "GENHLTH", "LSATISFY", "MARITAL","SSBSUGR2","SSBFRUT3"])
```

In [38]:
```
#Review data one last time
data.head()
```

Out[38]:

	Average_Number_of_Drinks_30_Days	BMI	Number_of_Children	Dentist_Visit_Last_Year
0	NaN	2249.0	0.0	1.0
1	NaN	2583.0	0.0	1.0
2	2.0	2253.0	0.0	0.0
3	NaN	2509.0	0.0	1.0
4	NaN	1977.0	0.0	1.0

5 rows × 32 columns

In [39]:

```
#Changing data type to categorical for necessary variables
data['Marital_Status']=pd.Categorical(data.Marital_Status)
data['Life_Satisfaction']=pd.Categorical(data.Life_Satisfaction)
data['General_Health_Personal_Rating']=pd.Categorical(data.General_Health_Personal_
data['Social_Emotional_Support']=pd.Categorical(data.Social_Emotional_Support)
data['Education']=pd.Categorical(data.Education)
data['Days_Poor_Physical_Health']=pd.Categorical(data.Days_Poor_Physical_Health)
data['Annual_Income']=pd.Categorical(data.Annual_Income)
data['Prediabetic_Diagnosis']=pd.Categorical(data.Prediabetic_Diagnosis)
data['Health_Rating']=pd.Categorical(data.Health_Rating)
data['Loneliness']=pd.Categorical(data.Loneliness)
data['Exercise_Outside_Work_Past_Month']=pd.Categorical(data.Exercise_Outside_Work_
data['Medical_Visit_Missed_Due_to_Cost']=pd.Categorical(data.Medical_Visit_Missed_D
data['Health_Insurance']=pd.Categorical(data.Health_Insurance)
data['Dentist_Visit_Last_Year']=pd.Categorical(data.Dentist_Visit_Last_Year)
```

In [40]:

```
#checking data types
data.dtypes
```

Out[40]:
```
Average_Number_of_Drinks_30_Days         float64
BMI                                      float64
Number_of_Children                       float64
Dentist_Visit_Last_Year                 category
Health_Insurance                        category
Marijuana_Use_30_Days                    float64
Medical_Visit_Missed_Due_to_Cost        category
Poor_Mental_Health_Days_Last_Month       float64
Poor_Physical_Health_Days_Last_Month     float64
Age_Category                            category
E_Cig_User                              category
Other_Tobacco_Usage                     category
Sex                                     category
Race                                    category
Time_Since_Last_Check_Up                category
Exercise_Outside_Work_Past_Month        category
Loneliness                              category
Health_Rating                           category
Prediabetic_Diagnosis                   category
Annual_Income                           category
Days_Poor_Physical_Health               category
Education                               category
Social_Emotional_Support                category
General_Health_Personal_Rating          category
Life_Satisfaction                       category
Marital_Status                          category
Sugar_Soda_Drinks_Per_Day                float64
Sugar_Soda_Drinks_Per_Week               float64
Sugar_Soda_Drinks_Per_Month              float64
Non_Soda_Sugar_Drinks_Per_Day            float64
Non_Soda_Sugar_Drinks_Per_Week           float64
Non_Soda_Sugar_Drinks_Per_Month          float64
dtype: object
```

In [41]:
```
#Output as .CSV
#Currently commented out since not always necessary to run
#data.to_csv("cleaned_data.csv")
```